# Portfolio selection with QAOA

Qiskit Fall Fest 2026 · Track 2 (Quantum Optimization) · Intermediate level.

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no quantum advantage was demonstrated (classical methods already reach the optimum).

Goal: pick the best set of assets with QAOA, then scale the problem up and stress it on hardware.

Everything runs on simulators; no IBM QPU job was submitted.

# Objective and cardinality constraint



- Buy exactly K of n assets: x_i = 1 means "buy asset i".
- Cost: C(x) = q * (x^T Sigma x) - (mu^T x) + lambda * (sum_i x_i - K)^2.
- Feasible set F = { x : sum_i x_i = K } (exactly-K portfolios).
- Instance (challenge, seed 2026, q = 0.5): n = 12, K = 6 -> 924 feasible portfolios.
- Encoding: x_i = (I - Z_i)/2; asset i maps to qubit i.
- Exact optimum: C = -1.066392092346649 2 at assets [1, 5, 6, 8, 10, 11].

# Encoding and the three quantum methods

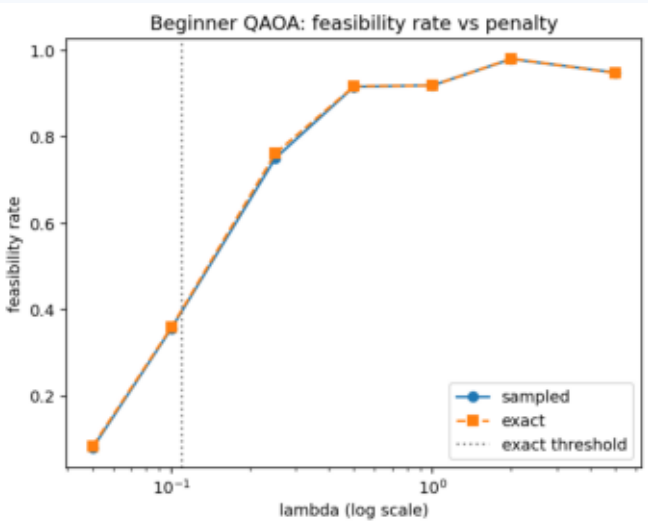


- Bit order: asset i <-> qubit i; state index s = sum_i x_i 2^i (qubit 0 = LSB).
- Method A: XY-ring mixer + uniform weight-K state, NO penalty -> every sample is feasible.
- Method B: warm start from a relaxation point, X-mixer, penalty lambda = 1.
- Method C: standard penalised QAOA, |+> initial state, penalty lambda = 1.
- Optimiser: COBYLA multistart; sampling is 4096 draws from the exact statevector, not hardware.

# Beginner baseline and penalty threshold



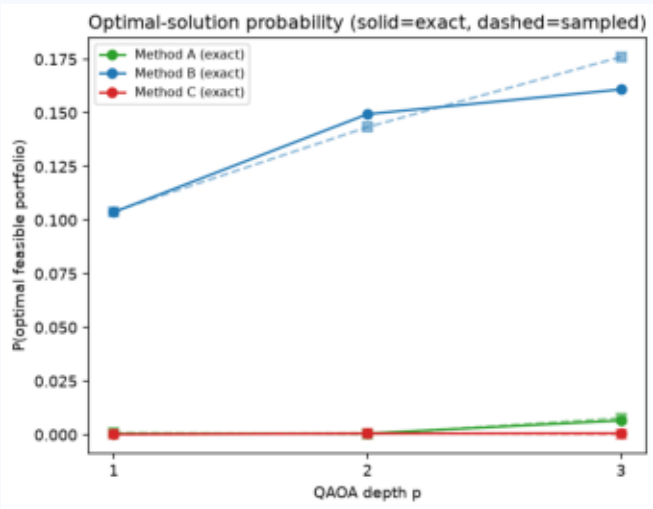
- Standard QAOA on n = 6, K = 3 (20 feasible portfolios).
- Exact penalty threshold: lambda* = 0.109661226.
- Below lambda*, QAOA's favourite answers break the budget; above it the feasible optimum is global.
- Feasibility rises with lambda (0.085 at lambda=0.05 -> ~0.98 at lambda=1).
- Best sampled feasible cost equals the exact optimum -0.503014690 at every depth.



# Intermediate results vs classical baselines



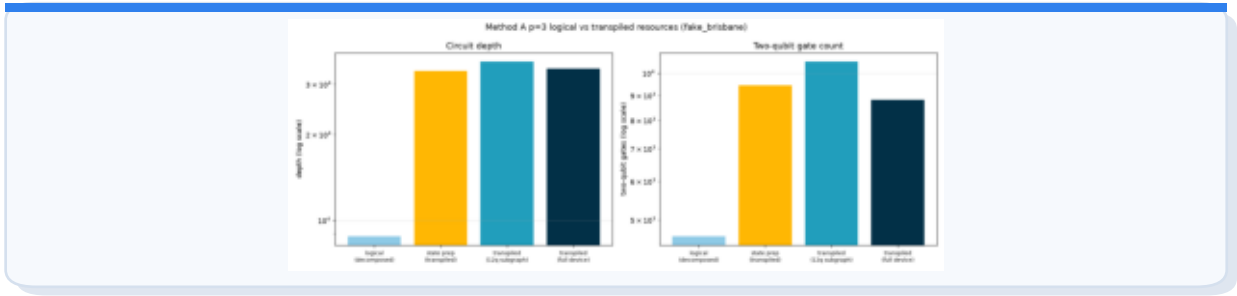
- Classical greedy AND feasible-swap SA both reach the exact optimum with gap 0 (under 2 s).
- Relaxation + rounding is also optimal; relaxation gives the lower bound -1.120144.
- Quantum A, B, C all tie on best feasible cost (-1.066392) - no advantage on quality.
- Probability of the optimum: B = 0.16, A = 0.0066, C = 0.0007.
- Intermediate exact penalty threshold: lambda* = 0.14936848561939664.



# Hardware transpilation and resource overhead



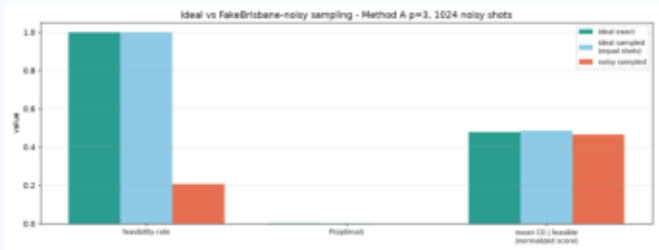
- Target: FakeBrisbane (127 qubits, 144 coupling edges, basis ecr/rz/sx/x).
- Logical Method A p = 3: depth 8782, 4623 two-qubit gates.
- Transpiled 12-qubit subgraph: depth 36008, 10554 ECR (4.10x depth, 2.28x two-qubit).
- Custom weight-K state prep dominates: 92.7% of depth, 89.3% of two-qubit gates.
- 51.4% of logical two-qubit pairs are non-adjacent -> routing shows up as extra ECR and depth.



# Trotterization gap vs hardware noise



- Gate-based circuit = first-order Trotter of the XY mixer -> an implementation gap, NOT noise.
- Trotter gap: TV = 0.5827 vs exact; P(opt) 0.0021 (circuit) vs 0.0066 (exact operator).
- FakeBrisbane noisy run (1024 shots): feasibility 1.0 -> 0.207, P(opt) -> 0.
- TV(exact ideal, noisy) = 0.8936; 908 of 4096 outcomes appeared, largest count only 3.
- At depth 36008 with 10554 ECR gates the circuit is far past the noise threshold.



# Conclusions, limitations and next steps



- No quantum advantage: classical greedy, SA and relaxation+rounding all reach the exact optimum.
- The constraint-preserving mixer guarantees feasibility but costs circuit depth.
- Two degradations kept separate: Trotterization (implementation) vs FakeBrisbane noise.
- Limitations: one synthetic instance; FakeBrisbane is a device model, not real hardware.
- Next: Advanced tier (n >= 20 with sector constraints), error mitigation, optional real QPU run.